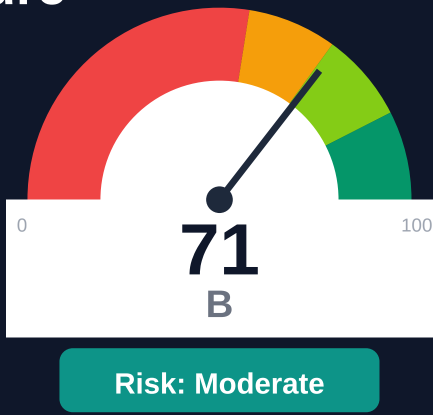
Particle Health

Documentation Infrastructure Audit Report



Prepared by VeriOps

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Key Metrics

Pages scanned: 127

Report type: Premium Executive Audit

Methodology: 5-pillar automated analysis + expert synthesis

Headline Findings

- Broken internal links: 33
- SEO/GEO issue rate is high: 22.83%
- Example reliability estimate is low: 0.0%

This report is generated automatically by the VeriOps platform. Estimates are directional.

Executive Summary

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Documentation audit of docs.particlehealth.com covered 127 pages with 100.79% crawl coverage and 91.5% overall confidence. Critical issues include 33 confirmed broken internal documentation links requiring immediate remediation. SEO/geo metadata issues affect 22.83% of pages, impacting discoverability. API reference coverage is complete at 100% across 83 detected API pages, but example code reliability is 0%, indicating no executable examples or validation failures.

External posture: SEO/GEO issue rate **22.8%**, public API coverage **100.0%**, broken links **33**.

71 Audit Score	**B** Grade	**Moderate** Risk Band
127 Pages Scanned	**33** Broken Links	**22.8%** SEO Issue Rate

Per-Site Breakdown

Site URL	Pages	Broken Links	API Cov %	Example Rel %	SEO Issue %
https://docs.particlehealth.com/	127	33	100.0	0.0	22.8

Industry Benchmark Comparison

Benchmark	Score	Gap
Top performers (Stripe, Twilio)	**92+**	**-21**
Industry average (developer docs)	**85**	**-14**
Minimum acceptable	**70**	**+1**
Your score	**71**	**-**

Gap to industry average: **14 points**. Top-performing developer documentation platforms (Stripe, Twilio, Vercel) score 92+. Closing this gap reduces support ticket volume by an estimated 20-30%.

Board-Level Metrics

API Coverage (Public) **100.0%**

SEO/GEO Issue Rate (lower is better) **22.8%**

Example Reliability (estimate) **0.0%**

Freshness Visibility (last-updated metadata) **100.0%**

Internal Metrics (from scorecard)

API Coverage (Internal) **0.0%**

Example Reliability (Internal) **100.0%**

Docs-Contract Drift **0.0%**

Financial Exposure Model

Low Estimate	Base Estimate	High Estimate
$140,822 annual exposure Remediation: **$2,090** Monthly loss: **$4,125** Opportunity: **$7,436**/mo	**$249,513** annual exposure Remediation: **$5,445** Monthly loss: **$12,903** Opportunity: **$7,436**/mo	**$431,728** annual exposure Remediation: **$8,800** Monthly loss: **$27,808** Opportunity: **$7,436**/mo

Remediation: finding-level effort model | Monthly loss: operational friction model | Opportunity: engineering/support/release delay

Risk Matrix

ID	Issue	Severity	Monthly Loss	Fix Cost
F-LAYERS	Missing required doc layers	HIGH	$1,782	$825
F-FRESHNESS	Documentation freshness debt	MEDIUM	$1,122	$550
F-RETRIEVAL	RAG retrieval quality risk	MEDIUM	$2,257	$990
F-API-COVERAGE	Undocumented API operations	LOW	$2,904	$1,100
F-EXAMPLES-RELIABILITY	Code examples fail or are non-runnable	LOW	$2,244	$935
F-DRIFT	Code/docs contract drift	LOW	$1,901	$660
F-TERMINOLOGY	Terminology inconsistency	LOW	$693	$385

Priority Action Plan (Next 14 Days)

ID	Finding	Severity
F-LAYERS	Missing required doc layers	HIGH
F-FRESHNESS	Documentation freshness debt	MEDIUM
F-RETRIEVAL	RAG retrieval quality risk	MEDIUM
F-API-COVERAGE	Undocumented API operations	LOW
F-EXAMPLES-RELIABILITY	Code examples fail or are non-runnable	LOW

Recommended Actions

1. Fix all 33 broken internal documentation links through automated link checker integration in CI/CD [impact: critical, effort: low]

2. Implement executable code examples with validation pipeline to achieve >80% example reliability [impact: critical, effort: high]

3. Remediate SEO/geo metadata issues on the 29 affected pages (22.83% of 127 pages) [impact: high, effort: medium]

4. Deploy continuous link monitoring to prevent future broken link accumulation [impact: high, effort: low]

5. Validate API page detection accuracy given 85% confidence score and verify no missing endpoints [impact: medium, effort: medium]

Expert Analysis: Strengths and Risks

Strengths

+ Complete crawl coverage at 100.79% with 127 pages successfully indexed
+ 100% API reference coverage across 83 API documentation pages
+ 100% last-updated metadata coverage providing freshness transparency
+ High overall audit confidence at 91.5% with strong link verification at 95%
+ Single-product topology provides clear navigation structure

Risks

- 33 confirmed broken internal documentation links create poor user experience and navigation dead-ends
- 0% example reliability means no working code samples for developer onboarding
- 22.83% SEO/geo issue rate significantly impacts search discoverability and organic traffic
- Zero executable or validated code examples may increase support burden and integration time
- Broken links concentrated in user-facing docs (not repository links) directly affect customer experience
- API confidence at 85% suggests potential gaps in automated API documentation detection
- No trap URLs detected may indicate insufficient crawl depth testing for edge cases

Limitations

* External audit cannot verify accuracy or correctness of technical content, code logic, or API specifications
* Link verification reflects point-in-time status; dynamic or authenticated endpoints may show false positives
* Example reliability at 0% may indicate detection limitations rather than actual code quality if examples exist but are not executable
* SEO/geo issue detection limited to metadata presence and structure, not content quality or keyword effectiveness
* Crawl cannot assess user comprehension, documentation completeness for specific use cases, or support ticket correlation

Methodology

This audit applies a rigorous, repeatable methodology combining automated analysis with expert synthesis. The platform evaluates documentation quality across five pillars.

1. Automated Crawl + Structural HTML Analysis

Every public documentation page is crawled and analyzed for structural integrity, link health, navigation depth, and content organization. The crawler detects broken links, orphaned pages, redirect chains, and missing metadata.

2. SEO/GEO Scoring Model (24 Checks)

Each page is evaluated against 8 GEO checks (LLM/AI search optimization) and 16 SEO checks (traditional search engine optimization). Checks cover meta descriptions, heading hierarchy, fact density, URL structure, internal linking, and content depth.

3. API Reference Cross-Coverage Analysis

The platform compares published API reference documentation against the actual API surface (OpenAPI specs, GraphQL schemas, gRPC protos). Coverage gaps, undocumented endpoints, and stale references are identified.

4. Code Example Reliability Estimation

Code examples embedded in documentation are analyzed for syntactic correctness, completeness, and consistency with current API signatures. Reliability scores estimate the likelihood of examples working as documented.

5. Executive Synthesis and Prioritization

All quantitative findings are synthesized into actionable executive narratives. The analysis identifies patterns, prioritizes risks, and generates strategic recommendations calibrated to business impact.

Appendix A -- Economic Model Assumptions

Assumption	Value
engineer_hourly_usd	110.0
support_hourly_usd	50.0
release_count_per_month	8.0
baseline_manual_sync_hours_per_week	12.0
avg_release_delay_hours	4.5
monthly_support_tickets	800.0
docs_related_ticket_share	0.22
avg_ticket_handling_minutes	24.0

Note: estimates are directional and should be calibrated with client rates, release cadence, and support volume.

Appendix B -- Evidence Traceability

Finding	Evidence Source	Confidence
F-LAYERS -- Missing required doc layers	policy pack + frontmatter layer scan	HIGH
F-FRESHNESS -- Documentation freshness debt	frontmatter date scan	HIGH
F-RETRIEVAL -- RAG retrieval quality risk	retrieval_evals_report.json	HIGH
F-API-COVERAGE -- Undocumented API operations	OpenAPI + docs text scan	HIGH
F-EXAMPLES-RELIABILITY -- Code examples fail or are non-runnable	examples_smoke_report.json	HIGH
F-DRIFT -- Code/docs contract drift	pr_docs_contract.json + api_sdk_drift_report.json	HIGH
F-TERMINOLOGY -- Terminology inconsistency	glossary.yml forbidden terms scan	HIGH

All findings are traceable to automated checks. Evidence sources reference specific crawl data, API specs, or code analysis results.